THREE HIGHER-ORDER IDENTITIES IN SUN'S CONJECTURE 3.5

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Abstract. We prove the two second-order identities and the third-order identity, namely (3.18)–(3.20), in Conjecture 3.5 of Zhi-Wei Sun. The proof uses one-parameter specializations of the Chu–Zhang acceleration of Dougall's nonterminating ${}_5F_4$ sum. For the second-order identities, an auxiliary parameter makes the reflected negative-index tail vanish and reduces the bilateral side to an exact sine quotient. For the third-order identity, a trace-free deformation forces the reflected tail to begin in degree three; its contribution is evaluated through

$$\sum_{k=0}^{\infty} \frac{H_k}{(2k+1)^2} = \frac{7\zeta(3) - \pi^2 \log 2}{4}.$$

The two appearances of $\pi^2 \log 2$ then cancel, leaving the conjectured multiple of $\zeta(3)$.

1. Statement

For $r \geq 1$, put

$$H_n^{(r)} := \sum_{j=1}^{n} \frac{1}{j^r}, \qquad H_0^{(r)} := 0,$$

and write $H_n = H_n^{(1)}$. Define

$$W_k := \frac{\binom{2k}{k}\binom{3k}{k}\binom{6k}{3k}}{(2k+1)4096^k}. \tag{1}$$

Sun's Conjecture 3.5 contains five identities built from this kernel [2, Conjecture 3.5]. The present note treats the three higher-order identities; the two first-order identities (3.16) and (3.17) are not discussed here.

Theorem 1.1. *The following identities hold:*

$$\sum_{k=0}^{\infty} W_k \left((74k+7)H_k^{(2)} + \frac{260(6k+1)}{(2k+1)^2} \right) = \frac{80}{3}\pi^2, \tag{2}$$

$$\sum_{k=0}^{\infty} W_k \left((74k+7)H_{2k}^{(2)} + \frac{205(6k+1)}{3(2k+1)^2} \right) = \frac{64}{9}\pi^2, \tag{3}$$

$$\sum_{k=0}^{\infty} W_k \left((74k+7)(9H_{2k}^{(3)} - H_k^{(3)}) - \frac{201(6k+1)}{(2k+1)^3} \right) = -160\zeta(3). \tag{4}$$

Thus equations (3.18)–(3.20) of Sun's Conjecture 3.5 are true.

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The underlying unweighted evaluation is Chu–Zhang's Example 60, which Sun records in the equivalent form

$$\sum_{k=0}^{\infty}(74k+7)W_k = 8. \tag{5}$$

The ratio of consecutive terms in all accelerated series below tends to $27/64$; consequently, all parameter differentiations used below are justified by normal convergence on sufficiently small compact parameter sets.

2. Hypergeometric input

For an integer n, let $(z)_n = \Gamma(z+n)/\Gamma(z)$ whenever the quotient is defined. Following Chu and Zhang [1], set

$$\Omega(a;b,c,d,e) := \sum_{k=0}^{\infty}(a+2k)\frac{(b)_k(c)_k(d)_k(e)_k}{(1+a-b)_k(1+a-c)_k(1+a-d)_k(1+a-e)_k} \tag{6}$$

and

$$\begin{aligned}\mathcal{G}(a;b,c,d,e) :=& \frac{\Gamma(1+a-b)\Gamma(1+a-c)\Gamma(1+a-d)\Gamma(1+a-e)}{\Gamma(b)\Gamma(c)\Gamma(d)\Gamma(e)}\\ &\times\frac{\Gamma(b+e-a)\Gamma(c+e-a)\Gamma(d+e-a)\Gamma(1+2a-b-c-d-e)}{\Gamma(1+a-b-c)\Gamma(1+a-b-d)\Gamma(1+a-c-d)}.\end{aligned} \tag{7}$$

Chu–Zhang's Theorem 27 states that, initially for $\Re(1+2a-b-c-d-e)>0$,

$$\begin{aligned}\Omega(a;b,c,d,e)-\mathcal{G}(a;b,c,d,e) =& \sum_{k=0}^{\infty}\frac{(e)_{3k}(b)_k(c)_k(d)_k}{(1+a-b)_{2k}(1+a-c)_{2k}(1+a-d)_{2k}}\\ &\times\frac{(1+a-b-c)_k(1+a-b-d)_k(1+a-c-d)_k}{(b+e-a)_k(c+e-a)_k(d+e-a)_k}\,\nu_k,\end{aligned} \tag{8}$$

where

$$\begin{aligned}\nu_k =& \frac{(a-c+2k)(a-e)}{a-c-e-k}\\ &-\frac{(c+k)(e+3k)(a-e)(1+a-b-d+k)}{(1+a-d+2k)(a-b-e-k)(a-c-e-k)}\\ &+\frac{(b+k)(c+k)(e+3k)(1+e+3k)(a-e)}{(1+a-b+2k)(1+a-c+2k)(1+a-d+2k)}\\ &\times\frac{(1+a-b-d+k)(1+a-c-d+k)}{(a-b-e-k)(a-c-e-k)(a-d-e-k)}.\end{aligned} \tag{9}$$

We shall also use the bilateral completion

$$\mathcal{B}(a;b,c,d,e) := \sum_{k\in\mathbb{Z}}(a+2k)\frac{(b)_k(c)_k(d)_k(e)_k}{(1+a-b)_k(1+a-c)_k(1+a-d)_k(1+a-e)_k}. \tag{10}$$

Dougall's bilateral ${}_5H_5$ summation, in this normalization, is

$$\mathcal{B}(a;b,c,d,e) = \frac{\Gamma(\Delta)\displaystyle\prod_{y\in\{b,c,d,e\}}\Gamma(1+a-y)\Gamma(1-y)}{\Gamma(a)\Gamma(1-a)\displaystyle\prod_{\{y,z\}\subset\{b,c,d,e\}}\Gamma(1+a-y-z)},\qquad \Delta = 1+2a-b-c-d-e. \tag{11}$$

For example, (11) follows from [3, (1.2b)] by setting $z = a/2$ and $g_y = a/2 - y$, and then multiplying by a; the latter converts $1 + k/z$ into $a + 2k$. It follows from Euler's reflection formula that

$$\mathcal{G}(a;b,c,d,e) = R(a;b,c,d,e)\mathcal{B}(a;b,c,d,e), \tag{12}$$

where

$$R(a;b,c,d,e) = \frac{\sin(\pi b)\sin(\pi c)\sin(\pi d)\sin(\pi e)}{\sin(\pi a)\sin\pi(b+e-a)\sin\pi(c+e-a)\sin\pi(d+e-a)}. \tag{13}$$

Let $\mathcal{N}$ denote the part of (10) with $k < 0$. Replacing k by $-m-1$ gives

$$\mathcal{N}(a;b,c,d,e) = -\kappa\,\Omega(2-a;1+b-a,1+c-a,1+d-a,1+e-a), \tag{14}$$

with

$$\kappa = \frac{(b-a)(c-a)(d-a)(e-a)}{(1-b)(1-c)(1-d)(1-e)}. \tag{15}$$

Finally, Chu–Zhang's reciprocal relation [1, Theorem 5] reads

$$\Omega(A;B,C,D,E) = \frac{A-E}{\Delta_*}\Omega(A_*;B_*,C_*,D_*,E), \tag{16}$$

where

$$\Delta_* = 1+2A-B-C-D-E, \qquad A_* = 1+2A-B-C-D,$$
$$B_* = 1+A-B-C, \qquad C_* = 1+A-B-D, \qquad D_* = 1+A-C-D.$$

All formulas in this section are first used where the relevant series converge and no denominator vanishes, and are then continued meromorphically.

3. The two second-order identities

Fix an auxiliary parameter x and specialize

$$a = c = \frac{1}{2}+t, \qquad b = \frac{1}{2}+(1+x)t, \qquad d = \frac{1}{2}+(1-x)t, \qquad e = \frac{1}{2}. \tag{17}$$

The balance is $-t$. Define the regularized difference

$$\Psi_x(t) := \frac{\Omega(a;b,c,d,e) - \mathcal{G}(a;b,c,d,e)}{t}. \tag{18}$$

The accelerated expression (8) shows directly that $\Psi_x(t)$ is holomorphic at $t = 0$.

3.1. **Exact evaluation of the bilateral side.** Since $c = a$, the negative-index terms vanish (equivalently, the factor $c-a$ in (15) is zero); indeed the corresponding denominator contains $(1)_{-m}$ for $m \geq 1$. Therefore $\Omega = \mathcal{B}$ on the path (17). Formula (11) simplifies to

$$\begin{aligned}\mathcal{B}_x(t) &:= \mathcal{B}(a;b,c,d,e)\\ &= \Gamma(-t)\frac{\Gamma(1-xt)\Gamma(1+xt)\Gamma(1+t)}{\Gamma(\frac{1}{2}+t)\Gamma(\frac{1}{2}-t)\Gamma(\frac{1}{2}-xt)\Gamma(\frac{1}{2}+xt)}\\ &= -xt\cot(\pi t)\cot(\pi xt). \end{aligned} \tag{19}$$

Moreover, (13) becomes

$$R_x(t) = \frac{\cos(\pi(1+x)t)\cos(\pi(1-x)t)}{\cos^2(\pi xt)}, \qquad 1 - R_x(t) = \frac{\sin^2(\pi t)}{\cos^2(\pi xt)}. \tag{20}$$

Since $\Omega - \mathcal{G} = (1-R_x)\mathcal{B}_x$, we obtain the exact identity

$$\boxed{\Psi_x(t) = -x\frac{\sin(2\pi t)}{\sin(2\pi xt)}.} \tag{21}$$

At $x=0$, the right-hand side is understood by continuity. In particular,

$$\Psi_x(t) = -1 + \frac{2\pi^2}{3}(1-x^2)t^2 + O(t^4), \qquad \Psi_x''(0) = \frac{4\pi^2}{3}(1-x^2). \tag{22}$$

3.2. **The accelerated side.** After substituting (17) into (8), write

$$\Psi_x(t) = \sum_{k=0}^{\infty} A_{k,x}(t) V_{k,x}(t), \tag{23}$$

where

$$\begin{aligned} A_{k,x}(t) = {} & \frac{(\frac12)_{3k}(\frac12+(1+x)t)_k(\frac12+t)_k(\frac12+(1-x)t)_k}{(1-xt)_{2k}(1)_{2k}(1+xt)_{2k}} \\ & \times \frac{(\frac12-(1+x)t)_k(\frac12-t)_k(\frac12-(1-x)t)_k}{(\frac12+xt)_k(\frac12)_k(\frac12-xt)_k}, \end{aligned} \tag{24}$$

and $V_{k,x}(t)$ is the specialization of ν_k/t. At $t=0$,

$$A_{k,x}(0) = Q_k := (2k+1)W_k. \tag{25}$$

For later use, recall the elementary formula

$$\frac{\mathrm{d}^r}{\mathrm{d}t^r} \log(\alpha+\lambda t)_m \bigg|_{t=0} = (-1)^{r-1}(r-1)!\lambda^r \sum_{j=0}^{m-1} \frac{1}{(\alpha+j)^r}. \tag{26}$$

Applying (26) to (24) gives

$$A_{k,x}'(0) = 0, \tag{27}$$

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2} \log A_{k,x}(t) \bigg|_{t=0} = 2(x^2+3)H_k^{(2)} - 6(x^2+4)H_{2k}^{(2)}. \tag{28}$$

Here we used

$$\sum_{j=0}^{k-1} \frac{1}{(j+\frac12)^2} = 4H_{2k}^{(2)} - H_k^{(2)}.$$

Direct substitution into the rational function (9) yields

$$V_{k,x}(0) = -\frac{74k+7}{8(2k+1)}, \tag{29}$$

$$V_{k,x}'(0) = 0, \tag{30}$$

$$V_{k,x}''(0) = -\frac{5(11x^2-8)(6k+1)}{4(2k+1)^3}. \tag{31}$$

An exact rational form for $V_{k,x}$ is recorded in Appendix A.

Differentiating (23) twice and using (25)–(31), we find

$$\begin{aligned} \Psi_x''(0) = {} & -\frac{x^2+3}{4}\sum_{k=0}^{\infty} W_k(74k+7)H_k^{(2)} \\ & + \frac{3(x^2+4)}{4}\sum_{k=0}^{\infty} W_k(74k+7)H_{2k}^{(2)} \\ & - \frac{5(11x^2-8)}{4}\sum_{k=0}^{\infty} W_k\frac{6k+1}{(2k+1)^2}. \end{aligned} \tag{32}$$

Set

$$X := \sum_{k=0}^{\infty} W_k(74k+7)H_k^{(2)},$$

$$Y := \sum_{k=0}^{\infty} W_k(74k+7)H_{2k}^{(2)},$$

$$Z := \sum_{k=0}^{\infty} W_k \frac{6k+1}{(2k+1)^2}.$$

Both sides are analytic in x near zero. Comparing the constant and x^2 coefficients in (22) and (32) gives

$$X - 3Y + 55Z = \frac{16}{3}\pi^2, \qquad -3X + 12Y + 40Z = \frac{16}{3}\pi^2. \tag{33}$$

Solving this system yields

$$X + 260Z = \frac{80}{3}\pi^2, \qquad Y + \frac{205}{3}Z = \frac{64}{9}\pi^2. \tag{34}$$

These are precisely (2) and (3).

4. The third-order identity

We now use the trace-free path

$$a = \frac{1}{2}, \qquad b = c = \frac{1}{2} + t, \qquad d = \frac{1}{2} - 2t, \qquad e = \frac{1}{2} - \varepsilon. \tag{35}$$

The balance is exactly ε. Define

$$\Phi(t) := \left.\frac{\Omega(a;b,c,d,e) - \mathcal{G}(a;b,c,d,e)}{\varepsilon}\right|_{\varepsilon=0}, \tag{36}$$

where the value at $\varepsilon = 0$ is taken through the normally convergent accelerated series (8).

4.1. **The accelerated third derivative.** At $\varepsilon = 0$, the Pochhammer product in (8) reduces to

$$A_k(t) = \frac{(\frac{1}{2})_{3k}(\frac{1}{2}-2t)_k(\frac{1}{2}+t)_k^2}{(1-t)_{2k}^2(1+2t)_{2k}}, \tag{37}$$

and we write $V_k(t)$ for the specialization of ν_k/ε at $\varepsilon = 0$. Thus

$$\Phi(t) = \sum_{k=0}^{\infty} A_k(t)V_k(t). \tag{38}$$

Let $\ell_k(t) = \log A_k(t)$. Formula (26) gives

$$\ell_k'(0) = 0, \qquad \ell_k'''(0) = -12\left(9H_{2k}^{(3)} - H_k^{(3)}\right). \tag{39}$$

The rational factor satisfies

$$V_k(0) = -\frac{74k+7}{8(2k+1)}, \tag{40}$$

$$V_k'(0) = 0, \tag{41}$$

$$V_k'''(0) = -\frac{603(6k+1)}{2(2k+1)^4}. \tag{42}$$

Since $A_k'(0) = V_k'(0) = 0$ and $A_k(0) = Q_k$, differentiating (38) three times gives

$$\Phi'''(0) = \frac{3}{2}\sum_{k=0}^{\infty} W_k \left((74k+7)(9H_{2k}^{(3)} - H_k^{(3)}) - \frac{201(6k+1)}{(2k+1)^3} \right). \tag{43}$$

It remains to prove

$$\Phi'''(0) = -240\zeta(3). \tag{44}$$

4.2. **Bilateral completion and the reflected tail.** On the path (35), formula (13) becomes

$$R(t,\varepsilon) = \frac{\cos^2(\pi t)\cos(2\pi t)\cos(\pi\varepsilon)}{\cos^2\big(\pi(t-\varepsilon)\big)\cos\big(\pi(2t+\varepsilon)\big)}. \tag{45}$$

Let

$$D(t,\varepsilon) := \varepsilon\mathcal{B}(t,\varepsilon), \tag{46}$$

where $\mathcal{B}(t,\varepsilon)$ denotes the corresponding specialization of (10). From (11), D is holomorphic at $\varepsilon = 0$, with

$$D_0(t) := D(t,0) = \frac{\Gamma(1-t)^2\Gamma(1+2t)}{\Gamma(\frac{1}{2})\Gamma(\frac{1}{2}+t)^2\Gamma(\frac{1}{2}-2t)}, \tag{47}$$

and

$$\frac{\partial_\varepsilon D(t,0)}{D_0(t)} = \delta(t) := 2\psi(1) + \psi\left(\frac{1}{2}\right) - 2\psi\left(\frac{1}{2}-t\right) - \psi\left(\frac{1}{2}+2t\right), \tag{48}$$

where $\psi = \Gamma'/\Gamma$.

The negative-tail formula (14), followed by the reciprocal relation (16), gives

$$\mathcal{N}(t,\varepsilon) = -\mathcal{C}(t)\mathcal{H}(t,\varepsilon), \tag{49}$$

where

$$\mathcal{C}(t) = \frac{2t^3}{(\frac{1}{2}-t)^2(\frac{1}{2}+2t)} \tag{50}$$

and

$$\mathcal{H}(t,\varepsilon) = \Omega\left(1; \frac{1}{2}-2t, \frac{1}{2}+t, \frac{1}{2}+t, 1-\varepsilon\right). \tag{51}$$

Indeed, before applying reciprocity the reflected series is

$$\Omega\left(\frac{3}{2}; 1+t, 1+t, 1-2t, 1-\varepsilon\right),$$

whose balance is ε; relation (16) transforms it into (51) with the factor $(\frac{1}{2}+\varepsilon)/\varepsilon$, which cancels the corresponding factors in (15).

Since $\mathcal{B} = \Omega + \mathcal{N}$ and $\mathcal{G} = R\mathcal{B}$, we have the exact decomposition

$$\Omega - \mathcal{G} = (1-R)\mathcal{B} + \mathcal{C}\mathcal{H}. \tag{52}$$

Put

$$\alpha(t) := \partial_\varepsilon R(t,0) = \pi\big(\tan(2\pi t) - 2\tan(\pi t)\big), \tag{53}$$

$$\beta(t) := \partial_\varepsilon^2 \log R(t,0) = \pi^2\big(2\sec^2(\pi t) + \sec^2(2\pi t) - 1\big). \tag{54}$$

Since $R(t,0) = 1$, one also has $\partial_\varepsilon^2 R(t,0) = \alpha(t)^2 + \beta(t)$. Let

$$\mathcal{H}_0(t) := \mathcal{H}(t,0), \qquad \mathcal{H}_1(t) := \partial_\varepsilon\mathcal{H}(t,0).$$

Dougall's nonterminating ${}_5F_4$ sum gives

$$\mathcal{H}_0(t) = \frac{\Gamma(\frac{3}{2}+2t)\Gamma(\frac{3}{2}-t)^2\Gamma(\frac{1}{2})}{\Gamma(1+t)^2\Gamma(1-2t)}. \tag{55}$$

Lemma 4.1 (Cancellation of the polar part)**.** *For t near zero,*

$$\alpha(t)D_0(t) = \mathcal{C}(t)\mathcal{H}_0(t). \tag{56}$$

Proof. Using $\Gamma(\frac{3}{2}-t) = (\frac{1}{2}-t)\Gamma(\frac{1}{2}-t)$ and $\Gamma(\frac{3}{2}+2t) = (\frac{1}{2}+2t)\Gamma(\frac{1}{2}+2t)$, followed by the reflection formula, one obtains

$$\frac{\mathcal{C}(t)\mathcal{H}_0(t)}{D_0(t)} = \pi\tan^2(\pi t)\tan(2\pi t).$$

The elementary identity

$$\tan(2z) - 2\tan z = \tan^2 z\tan(2z)$$

shows that the last expression is $\alpha(t)$. □

Expanding (52) at $\varepsilon = 0$, using $\mathcal{B} = D/\varepsilon$, gives

$$\frac{(1-R)\mathcal{B}+\mathcal{C}\mathcal{H}}{\varepsilon} = -\frac{\alpha D_0}{\varepsilon} - D_0\left(\alpha\delta + \frac{\alpha^2+\beta}{2}\right) + \frac{\mathcal{C}\mathcal{H}_0}{\varepsilon} + \mathcal{C}\mathcal{H}_1 + O(\varepsilon).$$

Lemma 4.1 cancels the polar terms. We therefore obtain the exact master identity

$$\boxed{\Phi(t) = -D_0(t)\left(\alpha(t)\delta(t) + \frac{\alpha(t)^2+\beta(t)}{2}\right) + \mathcal{C}(t)\mathcal{H}_1(t).} \tag{57}$$

Initially this identity holds in a punctured convergence region. Both sides are holomorphic near $t = 0$ (for $\mathcal{H}_1$, the balance is $1/2$), so the identity theorem extends it to a neighborhood of the origin.

4.3. **The cubic coefficient.** At $t = 0$, differentiating (51) termwise with respect to ε gives

$$\mathcal{H}_1(0) = -2\sum_{k=0}^{\infty}\frac{H_k}{(2k+1)^2}. \tag{58}$$

Lemma 4.2. *One has*

$$\sum_{k=0}^{\infty}\frac{H_k}{(2k+1)^2} = \frac{7\zeta(3)-\pi^2\log 2}{4}. \tag{59}$$

Consequently,

$$\mathcal{H}_1(0) = \frac{\pi^2\log 2 - 7\zeta(3)}{2}. \tag{60}$$

Proof. The generating function $\sum_{k\geq 0} H_k z^k = -\log(1-z)/(1-z)$ gives

$$\begin{aligned}\sum_{k=0}^{\infty}\frac{H_k}{(2k+1)^2} &= \int_0^1 \frac{\log x\,\log(1-x^2)}{1-x^2}\,\mathrm{d}x\\ &= \frac{1}{4}\int_0^1 t^{-1/2}\frac{\log t\,\log(1-t)}{1-t}\,\mathrm{d}t. \end{aligned}\tag{61}$$

Writing $\mathrm{B}(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, the last integral is

$$\frac{1}{4}\lim_{b\to 0^+}\partial_a\partial_b\mathrm{B}(a,b)\big|_{a=1/2}.$$

Write $\psi_m = \psi^{(m)}$. As $b \to 0$,

$$\partial_a \mathrm{B}(a,b) = -\psi_1(a) + b\left((\gamma + \psi(a))\psi_1(a) - \frac{1}{2}\psi_2(a)\right) + O(b^2).$$

Hence

$$\lim_{b\to 0^+} \partial_a\partial_b \mathrm{B}(a,b) = (\gamma + \psi(a))\psi_1(a) - \frac{1}{2}\psi_2(a).$$

Using

$$\psi\left(\frac{1}{2}\right) = -\gamma - 2\log 2, \qquad \psi_1\left(\frac{1}{2}\right) = \frac{\pi^2}{2}, \qquad \psi_2\left(\frac{1}{2}\right) = -14\zeta(3),$$

we obtain (59); equation (60) follows from (58). □

The required Taylor expansions are

$$D_0(t) = \frac{1}{\pi^2} - t^2 - \frac{16\zeta(3)}{\pi^2}t^3 + O(t^4), \tag{62}$$

$$\delta(t) = 4\log 2 + O(t^2), \tag{63}$$

$$\alpha(t) = 2\pi^4 t^3 + O(t^5), \tag{64}$$

$$\beta(t) = 2\pi^2 + 6\pi^4 t^2 + O(t^4), \tag{65}$$

$$\mathcal{C}(t) = 16t^3 + O(t^4), \qquad \mathcal{H}_1(t) = \mathcal{H}_1(0) + O(t). \tag{66}$$

Substitution into (57) gives, for the bilateral gamma part,

$$-D_0\left(\alpha\delta + \frac{\alpha^2+\beta}{2}\right) = -1 - 2\pi^2 t^2 + \bigl(16\zeta(3) - 8\pi^2\log 2\bigr)t^3 + O(t^4), \tag{67}$$

and, by (60), for the reflected tail,

$$\mathcal{C}(t)\mathcal{H}_1(t) = \bigl(8\pi^2\log 2 - 56\zeta(3)\bigr)t^3 + O(t^4). \tag{68}$$

The logarithmic terms cancel. Thus

$$\Phi(t) = -1 - 2\pi^2 t^2 - 40\zeta(3)t^3 + O(t^4), \tag{69}$$

so that $\Phi'''(0) = -240\zeta(3)$. Combining this with (43) proves (4) and completes the proof of Theorem 1.1.

Appendix A. The specialized rational factors

For completeness, we record exact rational expressions from which (29)–(31) and (40)–(42) follow by direct differentiation.

For the second-order path, put $n = 2k+1$. Then

$$V_{k,x}(t) = \frac{P_{n,x}(t)}{8n(n^2 - x^2t^2)(n^2 - 4x^2t^2)}, \tag{70}$$

where

$$\begin{aligned} P_{n,x}(t) = &-n^4(37n - 30) + 20n^2\bigl((n-2)x^2 + 6n - 4\bigr)t^2 \\ &+ 16\bigl(4(1-n)x^4 + (15n-10)x^2 + 6 - 9n\bigr)t^4. \end{aligned} \tag{71}$$

In particular, $V_{k,x}$ is even in t.

For the third-order path, again with $n = 2k+1$,

$$V_k(t) = \frac{37n^4 - 30n^3 + (-148n^3 + 120n^2)t + (384n^2 - 240n)t^2 + (-256n + 96)t^3 + 64t^4}{8(t-n)^2(4t-n)(n+2t)}. \tag{72}$$

These formulas are obtained by substituting the relevant paths directly into (9) before canceling the deformation parameter.

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